

CSE 381 Image Processing

Project Final Report

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Submitted

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Pipeline

Dataset traversal

The dataset is organized into subfolders corresponding to different puzzle sizes. The system iterates over each subfolder and processes every image independently.

Puzzle piece extraction

We slice the original image into different puzzle pieces

Image preprocessing

We apply canny edge detection to extract the edges which will be used later from matching

Pairwise compatibility computation

For every pair of puzzle pieces, directional compatibility scores are computed for all four possible adjacency directions. These scores measure how well two pieces fit together based on color continuity, gradient consistency, and edge alignment along their shared border.

Multi-start greedy assembly

Instead of one greedy run which can fall in local minima we run the greedy algorithm that finds the next best move multiple times to get the best score

Confidence scoring using best buddies

Each assembled puzzle is evaluated using a best-buddy consistency score, which measures how many adjacent piece pairs mutually select each other as their most compatible neighbor..

Best solution selection

Among all greedy runs, the solution with the highest best-buddy confidence score is selected as the final assembly.

Image reconstruction and saving

The selected piece arrangement is reconstructed into a full image by placing the corresponding puzzle pieces back into their grid positions.

Techniques

Puzzle Piece Extraction

Each image is split into equal-sized square pieces.

This guarantees perfect geometric alignment between pieces and avoids introducing rotation or scale ambiguity. Uniform pieces also allow compatibility to be computed using direct pixel comparisons.

Simplifies matching and ensures spatial consistency throughout the pipeline.

Image Preprocessing

Each image is split into equal-sized square pieces which guarantees perfect geometric alignment between pieces and avoids. Which is necessary for reordering the pieces

Also canny edge detection is applied to each piece to extract structural information. Edges capture object boundaries and line drawings, which were shown to be very important features when experimenting. This is particularly where large regions may have nearly identical colors. Also earlier color only matching failed in flat regions such as sky, smoke, or walls.

This adds a strong structural cue that is significant when measuring compatibility and placement scores later on.

Activity Scoring

Each piece receives an activity score based on the number of detected edges.

Pieces with high activity contain more distinctive features and are less ambiguous.

These pieces are preferred during seed selection and early assembly, because starting assembly from flat regions caused early locking and cascading errors.

This Increases stability by centering the solution around informative pieces, which lessens the chances of falling into a local minima.

Pairwise Compatibility Computation

Each pair of pieces is evaluated in all four possible directions.

Compatibility combines three cues:

- Color similarity along borders
- Gradient continuity across borders
- Edge alignment and continuity

Using multiple cues prevents one weak signal from dominating the decision. Color alone was insufficient for cartoons, while edges proved to be a strong feature. This produces reliable and discriminative matching scores which is the main driver for our solver

Best-Buddy Detection

Two pieces are considered best buddies if they mutually select each other as the best match in opposite directions.

This concept is directly taken from the reference paper.

Best buddies represent highly confident and symmetric relationships and asymmetric matches often lead to incorrect placements. This Provides a reliable indicator of correct adjacency which helps find the best alignment. Which across iterations will help get the best result for assembling.

Smart Seed Selection

The assembly process starts from a best-buddy pair that also has high activity. This avoids initializing the puzzle from ambiguous regions. The seed acts as the foundation for all future placements.

Incorrect seeds were observed to cause large-scale failure even when later matches were correct. This Improves global correctness by stabilizing the initial configuration which also helps in avoiding locking in on a wrong solution.

Greedy Local Assembly

Pieces are added one at a time to the current partial solution. For each empty slot, all possible pieces are evaluated against existing neighbors.

We choose the placement with the highest confidence which is how different the current solution is from the next best solution.

We chose this as fully global optimization is computationally expensive, while greedy assembly is efficient and effective when we combine correcting techniques.

This Builds the puzzle grid efficiently especially when repeated many times

Confidence Ratio Decision Rule

The confidence of a placement is measured using the ratio between the best and second-best match. A low ratio indicates a clear and reliable decision.

We chose this to delay ambiguous placements that ruin then the next placements as early experiments showed that accepting weak matches caused irreversible errors.

The result is preventing premature locking and improves overall stability.

Multi-Start Strategy

The greedy solver is executed multiple times. In some runs, small random noise is added to compatibility scores. This introduces controlled randomness and allows the solver to escape local minima.

This was introduced as single greedy run is sensitive to early mistakes so we add this term to the greedy algorithm to get different scores and choose the highest scores which got us more accurate results. This approach is inspired by global reasoning in the paper too

Increases robustness to initial errors which yields more accurate results the more we run but improvement decays the more we run the algorithm so the difference between running 10 times and 100 times is much more noticeable than running 100 times versus 1000 times.

Best-Buddy Confidence Scoring (Repeating best buddies)

Each completed solution is evaluated again by counting how many adjacent pairs are best buddies. Which of course ensures global consistency rather than local correctness. Solutions with many mutual matches are more likely to be correct overall so we always run the best buddies check after each iteration.

This Provides a reliable global quality measure that at the end decides what grid to choose.

Best Solution Selection

The solution with the highest best-buddy score is selected as the final output. This avoids relying on any single assembly attempt or simply running many times and getting the last result.

Ensures the final result is the most globally consistent solution and finally produces the best grid which is later produced in a directory.

Github Link:

<https://github.com/Knighler/Puzzles-Image-Processing>

Citation

Pomeranz, Dolev & Shemesh, Michal & Ben-Shahar, Ohad. (2011). A fully automated greedy square jigsaw puzzle solver. Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition. 9 - 16. 10.1109/CVPR.2011.5995331.